

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S6 (S, FE) / S6 (PT) (S, FE) Examination December 2024 (2019 Scheme)

Course Code: CST306**Course Name: ALGORITHM ANALYSIS AND DESIGN**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

- | | | |
|----|---|-----|
| 1 | Arrange the functions in the increasing order of their asymptotic growth rate
$2^n, n^{\log n}, n^2, n^{\sqrt{n}}$ | (3) |
| 2 | Explain the best case and worst case search operation in a binary search tree | (3) |
| 3 | Construct AVL tree for the following data 21,26,30,9,4,14,28 | (3) |
| 4 | Show the UNION operation using linked list representation of disjoint sets. | (3) |
| 5 | Give the control abstraction of Greedy strategy | (3) |
| 6 | Compare Strassen's matrix multiplication with ordinary matrix multiplication | (3) |
| 7 | Compare backtracking with branch and bound technique | (3) |
| 8 | Illustrate how optimal substructure property could be maintained in Floyd Warshall algorithm. | (3) |
| 9 | Explain P, NP and NP Hard Problems with examples | (3) |
| 10 | Compare Monte Carlo and Las Vegas algorithms | (3) |

PART B*Answer one full question from each module, each carries 14 marks.***Module I**

- | | | |
|----|---|-----|
| 11 | a) Find the time complexity of the following codes | (6) |
| | <pre> (i) int fun(int n){ int i, j; for(i=0; i<n; i++) { for(j=0; j<i; j++){ printf("%d%d",i,j); } } } </pre> | |

```
(ii) sum=0;
for(i=0; i<n; i++)
for (j=1; j<n; j= j*2)
sum++ ;
```

- b) State and Explain masters theorem. Using masters theorem find the complexity of given functions (8)

- (i) $T(n) = 2T(n/2) + n \log n$
(ii) $T(n) = 64T(n/8) - n^2 \log n$

OR

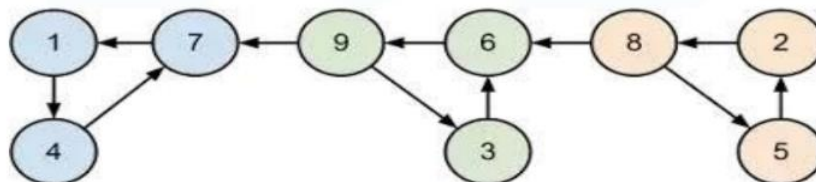
- 12 a) Using recursion tree method, solve the recurrence equations (8)

- (i) $T(n) = T(n/5) + T(4n/5) + n$
(ii) $T(n) = 3T(n/4) + cn^2$

- b) Explain with examples, different notations used in asymptotic analysis (6)

Module II

- 13 a) What are strongly connected components of a graph, Explain the algorithm to find the SCC. Also find the strongly connected components of a given graph (8)



- b) Explain DFS algorithm of a graph. Also find the time complexity of the algorithm (6)

OR

- 14 a) What are the rules to be followed while deleting an element from AVL trees? (8)
Construct an AVL trees with the following data: 44, 17, 32, 78, 50, 88, 48, 62, 84, 92, 80, 82. Also delete the elements 62, 84, 78, 17. Explain each steps.

- b) Explain BFS algorithm of a graph. Also find the time complexity of the algorithm (6)

Module III

- 15 a) Explain 2- way merge sort algorithm with an example and derive its time complexity. (7)

b) Given Objects:

(7)

	1	2	3	4	5	6	7
Profit (P):	5	10	15	7	8	9	4
Weight(w):	1	3	5	4	1	3	2

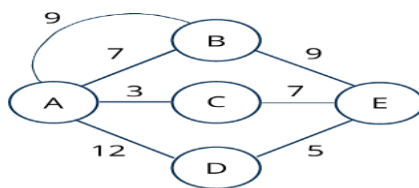
W (Weight of the knapsack): 15

Find the optimal solution for the following Fractional Knapsack problem.

OR

16 a) Write Dijkstra's algorithm for single source shortest path. Perform its complexity analysis. (7)

b) Find the cost of Minimal Spanning Tree of the given graph by using Kruskal's Algorithm (7)



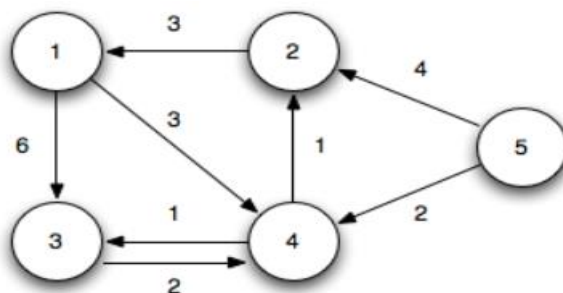
Module IV

17 a) Explain the concept of Backtracking method using 4 Queens problem. (6)

b) Given the chain of 4 matrices A1, A2, A3, A4 with sequence of dimensions 5x6, 6x4, 4x2, 2x3 . Using Matrix chain multiplication, find the optimal paranthesis of the given problem. (8)

OR

18 a) Using Floyd Warshall Algorithm, find the shortest path distance between every pair of vertices. (8)



b) Explain how Travelling Salesman Problem can be solved using Branch and Bound method. (6)

Module V

- 19 a) Prove that Clique problem is NP Hard (7)
 b) Define Bin Packing problem. Give the First Fit heuristic approximation for this (7)
 problem. What is the approximation ratio of this heuristic?

OR

- 20 a) Prove that vertex cover problem is NP Complete (7)
 b) Explain randomized quick sort with the help of suitable examples (7)

Course Code: CST306

Course Name: ALGORITHM ANALYSIS AND DESIGN

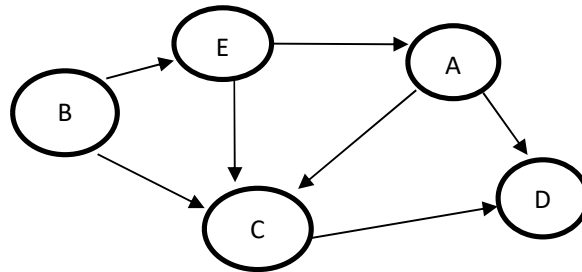
Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

- | | | |
|---|---|-----|
| 1 | Explain the best case and worst case of a search operation in a binary search tree. | (3) |
| 2 | Is $n^2 \in o(n^2)$? (Little o). Justify your answer. | (3) |
| 3 | Explain about any two cases of rotations in an AVL-Tree with proper example. | (3) |
| 4 | Obtain the topological sort order of the given graph $G=(V,E)$ | (3) |



- | | | |
|----|---|-----|
| 5 | Give the control abstraction of divide and conquer algorithm design strategy. | (3) |
| 6 | Consider an instance of fractional knapsack problem with 3 objects and capacity 20Kg. The profit and weight vectors corresponding to the 3 objects respectively are (10, 15, 20) and (8, 10, 20). Compute an optimal solution to this instance of fractional knapsack problem. What will be the profit earned corresponding to this solution? | (3) |
| 7 | What are the main steps involved in solving a computational problem using the algorithm design strategy Dynamic Programming. | (3) |
| 8 | List any three differences between Backtracking and Branch and Bound strategies. | (3) |
| 9 | Let $G = (V, E)$ be a graph with vertex set V and edge set E . Define the optimization and decision versions of CLIQUE problem in G . | (3) |
| 10 | Compare Las Vegas Algorithm and Monte Carlo algorithms. | (3) |

PART B*Answer one full question from each module, each carries 14 marks.***Module I**

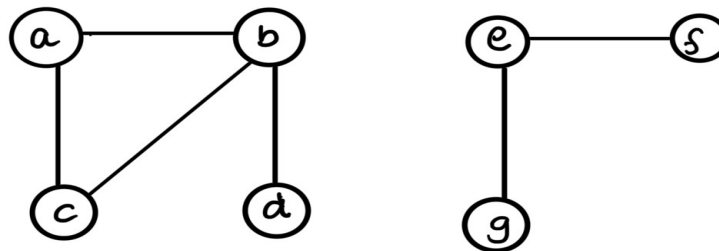
- 11 a) Illustrate the best case, worst case and average case of linear search algorithm with proper examples. (7)
- b) Solve the following recurrence using recursion tree method. (7)
- $$T(n) = 2T(n/2) + 1$$

OR

- 12 a) Solve the following recurrence using substitution method. (8)
- $$T(n) = 3T\left(\frac{n}{4}\right) + cn^2$$
- b) Solve the following recurrences using Master's theorem. (6)
- i) $T(n) = 2T\left(\frac{n}{4}\right) + 1$
- ii) $T(n) = 2T\left(\frac{n}{4}\right) + \sqrt{n}$

Module II

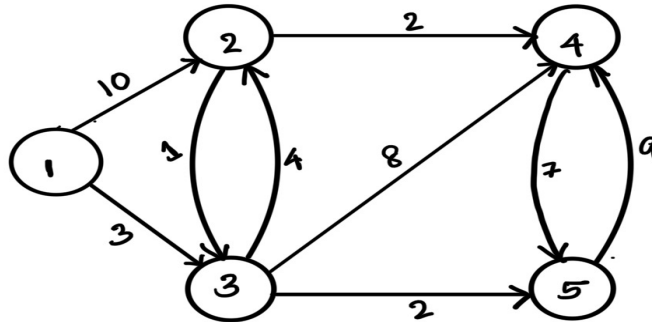
- 13 a) Create an AVL tree by inserting the values from 1 to 8 in increasing order. (7)
- b) Give an algorithm for computing connected components in a graph using Disjoint Set data structure operations. Explain the working of this algorithm on the graph given below. (7)

**OR**

- 14 a) Give an appropriate algorithm for Depth First Search in a given connected graph. Derive the time complexity of the algorithm with proper justification. (9)
- b) What are strongly connected components of a directed graph? Explain with an example. (5)

Module III

- 15 a) Apply Dijkstra's algorithm on the given graph to find the shortest path to all destinations starting from vertex with label 1 (6)



- b) Write Kruskal's algorithm to find the minimum cost spanning tree of a given weighted connected graph. What is the time complexity of the algorithm? Justify your answer. (8)

OR

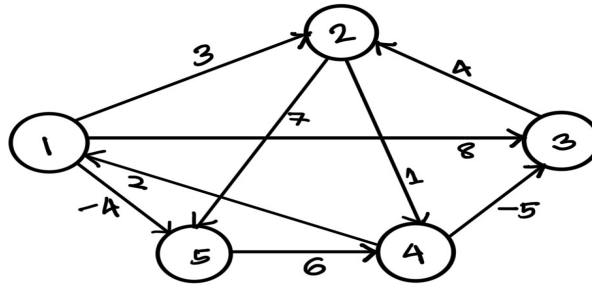
- 16 a) Formally state the fractional knapsack problem. Give an algorithm for computing an optimal solution to this problem using greedy strategy. What is its time complexity? Justify your answer. (9)
- b) Give the control abstraction of greedy algorithm design strategy. (5)

Module IV

- 17 a) Given a chain of 4 matrices $\langle A_1, A_2, A_3, A_4 \rangle$ with dimensions $\langle 5 \times 4 \rangle, \langle 4 \times 6 \rangle, \langle 6 \times 2 \rangle$, and $\langle 2 \times 7 \rangle$. Fully parenthesize the given matrices so that the number of scalar multiplications required to compute their product is minimum. What is the value of optimum number of scalar multiplications to compute the product of the given chain of matrices? (9)
- b) Define the control abstraction for Backtracking (5)

OR

- 18 a) Consider the weighted directed graph given in the following Figure. (8)



Construct the weight adjacency matrix of the given graph. Apply the Floyd-Warshall algorithm to construct the matrix D^2 that represents the shortest paths distance between all vertices i and j ($1 \leq i \leq 5$ and $1 \leq j \leq 5$) through intermediate vertices 1 and 2.

- b) Define n-queens problem. Draw the state space diagram of 4-queens problem by applying backtrack method. (6)

Module V

- 19 a) Let $G = (V, E)$ be a graph with vertex set V and edge set E . Prove that the decision version of Clique problem in G belongs to the complexity class NP. (5)
- b) Give the randomized version of Quick Sort and prove that the expected number of comparisons performed by this algorithm is $O(n \log_2 n)$ (9)

OR

- 20 a) Let $G = (V, E)$ be a graph with vertex set V and edge set E . What is meant by a vertex cover of G ? Also prove that the decision version of vertex cover problem is NP-Hard. (8)
- b) Define Bin Packing problem. Give the First Fit heuristic approximation for this problem. What is the approximation ratio of this heuristic? (6)

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Sixth Semester B.Tech Degree Supplementary Examination May 2023 (2019 Scheme)

Course Code: CST306**Course Name: ALGORITHM ANALYSIS AND DESIGN**

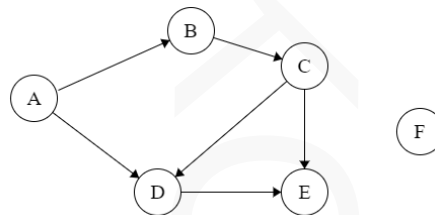
Max. Marks: 100

Duration: 3 Hours

PART A**Answer all questions, each carries 3 marks.**

Marks

- 1 Let $f(n) = 7n + 4$. Prove that this is of the order of $\Omega(n)$. (3)
- 2 Solve the following recurrence using Master theorem. (3)
 - a) $T(n) = 8T\left(\frac{n}{2}\right) + n^2$
 - b) $T(n) = 2T\left(\frac{n}{2}\right) + n$
- 3 Define MAKE_SET(x), UNION(x,y) and FIND_SET(x) operations of disjoint set data structure with a suitable example. (3)
- 4 Find any ONE topological ordering of the following the graph (3)



- 5 Give the control abstraction of Divide and Conquer algorithm design strategy. (3)
- 6 Apply greedy algorithm for fractional knapsack to find the optimal ordering for loading the items in the knapsack. Let the knapsack capacity, $M=15$ (3)

Items	Weight	Profit
1	8	24
2	9	18
3	5	20

- 7 Explain about the structure of an optimal paranthesization of matrix-chain multiplication problem. (3)
- 8 Distinguish the branch-and-bound technique from the backtracking technique. (3)
- 9 Discuss the need for approximation algorithm. (3)
- 10 Define graph colouring problem. (3)

PART B*Answer one full question from each module, each carries 14 marks.***Module I**

- 11 a) Define the asymptotic notations: Big Oh, Big Omega, Theta, little oh and little omega. (6)

- b) Solve the following recurrence using recursion tree method: (8)

1) $T(n) = T\left(\frac{n}{2}\right) + 1, T(1) = 1$

2) $T(n) = 2T\left(\frac{n}{2}\right) + n^2, T(1) = 1$

OR

- 12 a) Perform complexity analysis for the following code segments: (6)

```

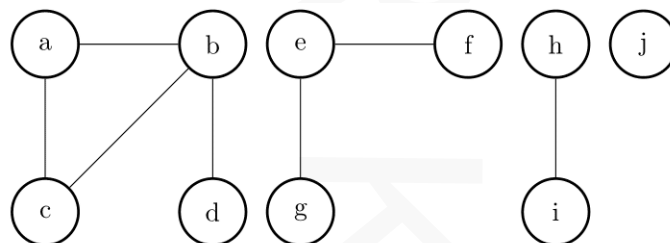
1) For i=1 to n do
    For j = 1 to n do
        A=B * C
    End for
End for
2) Function F(n)
    {
        If(n==0)
            return(1)
        Else
            return (n*F(n-1))
        }

```

- b) Discuss the concept of best case, worst case and average case complexity of an algorithm with linear search algorithm. (8)

Module II

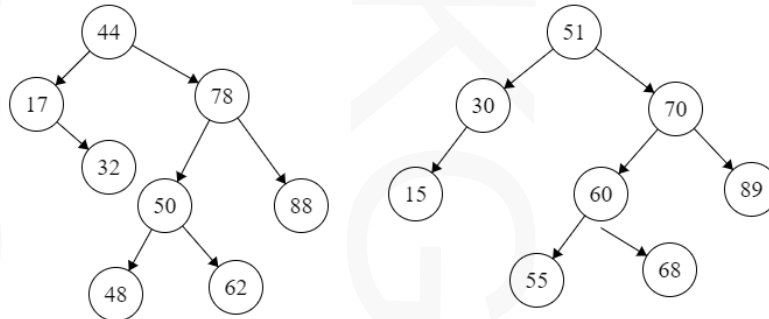
- 13 a) What are the operations supported by Disjoint Data Structure? Explain the working of Disjoint Set Data Structure for computing Connected Components of an undirected graph given in the following Figure. (7)



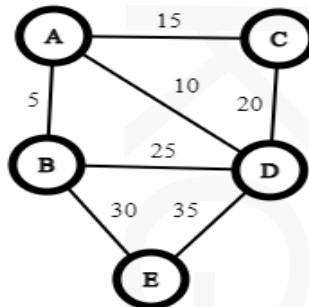
- b) Illustrate the advantage of AVL trees with a suitable example. Discuss the various rotations required to balance the height of AVL tree during insertion and deletion. (7)

OR

- 14 a) Give BFS algorithm for graph traversal and perform its complexity analysis. (7)
 b) Perform the following operations in the given AVL trees. (7)
 1) Insert 54 in Tree 1 2) Delete 15 from Tree 2.

**Module III**

- 15 a) Write a divide and conquer algorithm for 2-way merge sort and perform its complexity analysis. (7)
 b) Give Kruskal's algorithm for minimum cost spanning tree computation. Apply the algorithm to find the minimum cost spanning tree for the following graph (7)



OR

- 16 a) Write Dijkstra's algorithm for single source shortest path. Perform its complexity analysis. (7)
 b) Write the control abstraction for Greedy design technique. Give a greedy algorithm for fractional knapsack problem. (7)

Module IV

- 17 a) Write Floyd Warshall algorithm for all pair shortest path problem and perform its complexity analysis. (5)

- b) Define Travelling Salesman Problem. Solve the following instance of TSP using branch and bound technique. The cost matrix is given below: (9)

	A	B	C	D	E
A	-	10	8	9	7
B	10	-	10	5	6
C	8	10	-	8	9
D	9	5	8	-	6
E	7	6	9	6	-

OR

- 18 a) Consider the following four matrices and perform chain matrix multiplication using the dynamic programming approach. Finally give the optimal cost of multiplication and optimal parenthesization. (8)

A	B	C	D
4x5	5x3	3x2	2x7

- b) Define n-queens problem. Draw the state space tree for 4-queens problem using backtracking method. (6)

Module V

- 19 a) Explain the benefits of randomized algorithm over deterministic algorithm. Discuss briefly the major categories of randomized algorithms. Give example for each category. (7)
- b) Define bin packing problem. Discuss the first fit strategy for solving it. State the approximation ratio of the algorithm. (7)

OR

- 20 a) Give a randomized version of quicksort algorithm and perform its expected running time analysis. (7)
- b) Prove that vertex cover problem is NP Complete. (7)

Course Code: CST306

Course Name: ALGORITHM ANALYSIS AND DESIGN

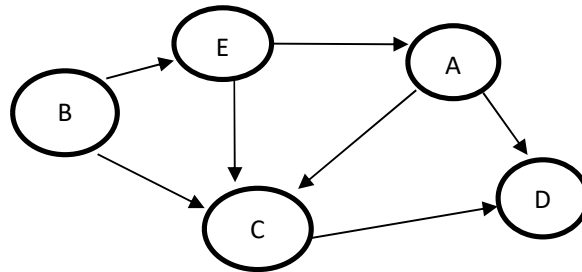
Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

- | | | |
|---|---|-----|
| 1 | Explain the best case and worst case of a search operation in a binary search tree. | (3) |
| 2 | Is $n^2 \in o(n^2)$? (Little o). Justify your answer. | (3) |
| 3 | Explain about any two cases of rotations in an AVL-Tree with proper example. | (3) |
| 4 | Obtain the topological sort order of the given graph $G=(V,E)$ | (3) |



- | | | |
|----|---|-----|
| 5 | Give the control abstraction of divide and conquer algorithm design strategy. | (3) |
| 6 | Consider an instance of fractional knapsack problem with 3 objects and capacity 20Kg. The profit and weight vectors corresponding to the 3 objects respectively are (10, 15, 20) and (8, 10, 20). Compute an optimal solution to this instance of fractional knapsack problem. What will be the profit earned corresponding to this solution? | (3) |
| 7 | What are the main steps involved in solving a computational problem using the algorithm design strategy Dynamic Programming. | (3) |
| 8 | List any three differences between Backtracking and Branch and Bound strategies. | (3) |
| 9 | Let $G = (V, E)$ be a graph with vertex set V and edge set E . Define the optimization and decision versions of CLIQUE problem in G . | (3) |
| 10 | Compare Las Vegas Algorithm and Monte Carlo algorithms. | (3) |

PART B*Answer one full question from each module, each carries 14 marks.***Module I**

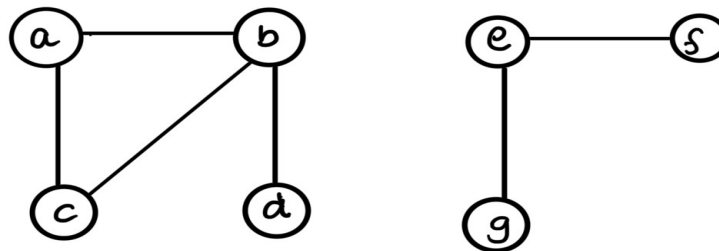
- 11 a) Illustrate the best case, worst case and average case of linear search algorithm with proper examples. (7)
- b) Solve the following recurrence using recursion tree method. (7)
- $$T(n) = 2T(n/2) + 1$$

OR

- 12 a) Solve the following recurrence using substitution method. (8)
- $$T(n) = 3T\left(\frac{n}{4}\right) + cn^2$$
- b) Solve the following recurrences using Master's theorem. (6)
- i) $T(n) = 2T\left(\frac{n}{4}\right) + 1$
- ii) $T(n) = 2T\left(\frac{n}{4}\right) + \sqrt{n}$

Module II

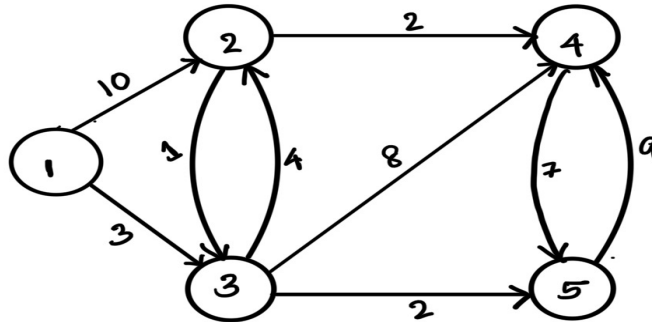
- 13 a) Create an AVL tree by inserting the values from 1 to 8 in increasing order. (7)
- b) Give an algorithm for computing connected components in a graph using Disjoint Set data structure operations. Explain the working of this algorithm on the graph given below. (7)

**OR**

- 14 a) Give an appropriate algorithm for Depth First Search in a given connected graph. Derive the time complexity of the algorithm with proper justification. (9)
- b) What are strongly connected components of a directed graph? Explain with an example. (5)

Module III

- 15 a) Apply Dijkstra's algorithm on the given graph to find the shortest path to all destinations starting from vertex with label 1 (6)



- b) Write Kruskal's algorithm to find the minimum cost spanning tree of a given weighted connected graph. What is the time complexity of the algorithm? Justify your answer. (8)

OR

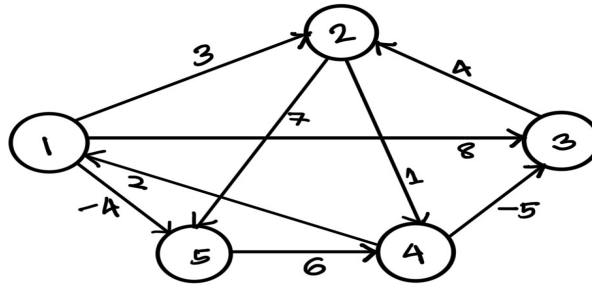
- 16 a) Formally state the fractional knapsack problem. Give an algorithm for computing an optimal solution to this problem using greedy strategy. What is its time complexity? Justify your answer. (9)
- b) Give the control abstraction of greedy algorithm design strategy. (5)

Module IV

- 17 a) Given a chain of 4 matrices $\langle A_1, A_2, A_3, A_4 \rangle$ with dimensions $\langle 5 \times 4 \rangle, \langle 4 \times 6 \rangle, \langle 6 \times 2 \rangle$, and $\langle 2 \times 7 \rangle$. Fully parenthesize the given matrices so that the number of scalar multiplications required to compute their product is minimum. What is the value of optimum number of scalar multiplications to compute the product of the given chain of matrices? (9)
- b) Define the control abstraction for Backtracking (5)

OR

- 18 a) Consider the weighted directed graph given in the following Figure. (8)



Construct the weight adjacency matrix of the given graph. Apply the Floyd-Warshall algorithm to construct the matrix D^2 that represents the shortest paths distance between all vertices i and j ($1 \leq i \leq 5$ and $1 \leq j \leq 5$) through intermediate vertices 1 and 2.

- b) Define n-queens problem. Draw the state space diagram of 4-queens problem by applying backtrack method. (6)

Module V

- 19 a) Let $G = (V, E)$ be a graph with vertex set V and edge set E . Prove that the decision version of Clique problem in G belongs to the complexity class NP. (5)
- b) Give the randomized version of Quick Sort and prove that the expected number of comparisons performed by this algorithm is $O(n \log_2 n)$ (9)

OR

- 20 a) Let $G = (V, E)$ be a graph with vertex set V and edge set E . What is meant by a vertex cover of G ? Also prove that the decision version of vertex cover problem is NP-Hard. (8)
- b) Define Bin Packing problem. Give the First Fit heuristic approximation for this problem. What is the approximation ratio of this heuristic? (6)

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S6 (R, S) / S6 (PT) (R) Examination June 2023 (2019 Scheme)

Course Code: CST306**Course Name: ALGORITHM ANALYSIS AND DESIGN**

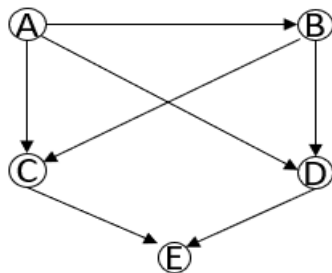
Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

- 1 Show that for any real constants a and b , where $b > 0$, $(n + a)^b = O(n^b)$ (3)
- 2 Solve the following recurrence equations using Master theorem.
 a. $T(n) = 3T(n/2) + n^2$ (3)
 b. $T(n) = 2T(n/2) + n \log n$
- 3 Define AVL tree. Explain the rotations performed for insertion in AVL tree. (3)
- 4 Find the different topological ordering of the given graph. (3)



- 5 Write the control abstraction of divide and conquer strategy. (3)
- 6 Compare Strassen's matrix multiplication with ordinary matrix multiplication. (3)
- 7 Differentiate backtracking technique from branch and bound technique. (3)
- 8 What is Principle of Optimality? (3)
- 9 Differentiate P and NP problems. Give one example to each. (3)
- 10 Define graph coloring problem. (3)

PART B*Answer one full question from each module, each carries 14 marks.***Module I**

- 11 a) Define Big Oh, Big Omega and Theta notations and illustrate them graphically. (7)
- b) Find the time complexity of following code segment (7)


```

      (i) for (int i = 1; i <= n; i *= c) {
          // some O(1) expressions
      }
      
```

```

    for (int i = n; i > 0; i /= c) {
        // some O(1) expressions
    }
(ii) for (int i = 1; i <= n; i += c) {
    // some O(1) expressions
}
for (int i = n; i > 0; i -= c) {
    // some O(1) expressions
}

```

OR

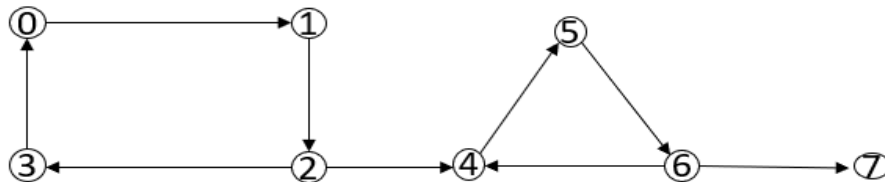
- 12 a) Find the best case, worst case and average case time complexity of binary search. (7)
- b) Find the time complexity of following function using recursion tree method. (7)
- (i) $T(n) = 2 T(n/2) + n^2$
- (ii) $T(n) = T(n/3) + T(2n/3) + n$

Module II

- 13 a) Construct AVL tree by inserting following elements appeared in the order. (7)
21, 26, 30, 9, 4, 14, 28, 18, 15
- b) Explain union and find algorithms in disjoint datasets. (7)

OR

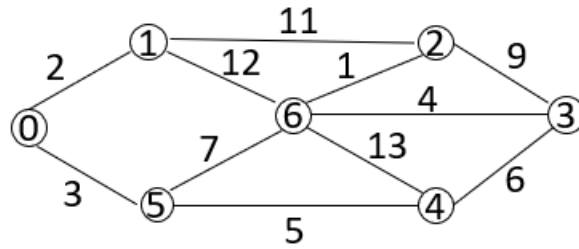
- 14 a) Write DFS algorithm for graph traversal. Also derive its time complexity. (7)
- b) Find the strongly connected components of the given directed graph. (7)

**Module III**

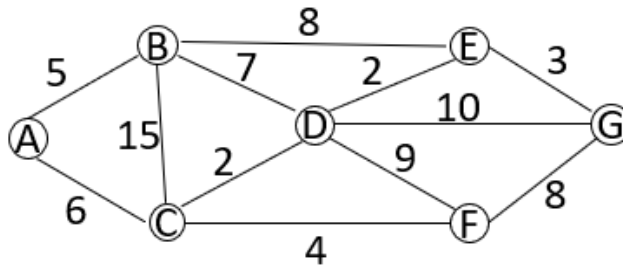
- 15 a) Explain 2- way merge sort algorithm with an example and derive its time complexity. (7)
- b) Find the optimal solution for the following Fractional Knapsack problem. (7)
Given the number of items(n) = 7, capacity of sack(m) = 15,
 $W = \{1, 3, 5, 4, 1, 3, 2\}$ and $P = \{10, 15, 7, 8, 9, 4\}$

OR

- 16 a) Apply Kruskal's algorithm for finding minimum cost spanning tree. (7)



- b) Apply Dijkstra's algorithm for finding the shortest path from vertex A to all other vertices. (7)

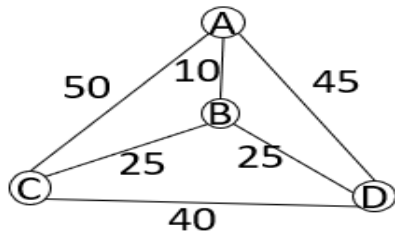


Module IV

- 17 a) Find the optimal parenthesis of matrix chain product whose sequence of dimensions is $5 \times 4, 4 \times 6, 6 \times 2, 2 \times 7$ (8)
 b) Explain 4 queen problem. Draw the state space tree for 4 queen problem. (6)

OR

- 18 a) Define TSP problem. Apply branch and bound algorithm for solving TSP. (9)



- b) Write Floyd Warshall's algorithm for finding all pairs shortest path algorithm. (5)

Module V

- 19 a) Explain the first fit-decreasing strategy of bin packing algorithm. (7)
 b) Prove that Clique Decision problem is NP-complete. (7)

OR

- 20 a) Differentiate Las Vegas and Monte Carlo algorithms (7)
 b) Explain randomized quick sort with the help of suitable examples. (7)

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Sixth Semester B.Tech Degree Examination June 2022 (2019 Scheme)

Course Code: CST306**Course Name: ALGORITHM ANALYSIS AND DESIGN**

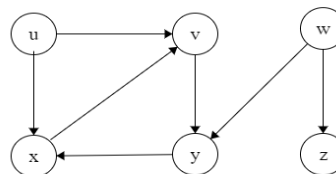
Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

- 1 Let $f(n) = 3n^3 + 2n^2 + 3$ for an algorithm, Let $g(n) = n^3$. Prove that $f(n)$ of this algorithm is in $O(n^3)$ (3)
- 2 Solve the recurrence $T(n) = 3T\left(\frac{n}{4}\right) + n \log n$. using Master theorem. (3)
- 3 Discuss briefly the heuristics, union by rank and path compression, to improve the running time of disjoint set data structure. (3)
- 4 Consider the directed acyclic graph $G=(V,E)$ given in the following figure. (3)



Find any topological ordering of G.

- 5 Give the control abstraction of Greedy strategy. (3)
- 6 Why Strassen's matrix multiplication algorithm is better than traditional divide and conquer algorithm for multiplying two square matrices? What is the recurrence for the number of computational steps taken by Strassen's algorithm and its time complexity? (3)
- 7 Discuss briefly the elements of dynamic programming with a suitable example. (3)
- 8 Compare backtracking and branch-and-bound design techniques. (3)
- 9 Define P, NP and NP complete domains. (3)
- 10 Compare Las Vegas and Monte Carlo algorithms. (3)

PART B*Answer one full question from each module, each carries 14 marks.***Module I**

- 11 a) Illustrate best case, average case and worst-case complexity with insertion sort algorithm. (9)

- b) Give the general idea of the substitution method for solving recurrences. Solve the following recurrence using substitution method. (5)

$$T(n) = 2T\left(\frac{n}{2}\right) + n$$

OR

- 12 a) Solve the following recurrence using recursion tree method (8)

$$1) T(n) = T\left(\frac{n}{3}\right) + T\left(\frac{2n}{3}\right) + cn$$

$$2) T(n) = 2T\left(\frac{n}{2}\right) + n$$

- b) Define Big Oh, Big Omega and Theta notations and illustrate them graphically. (6)

Module II

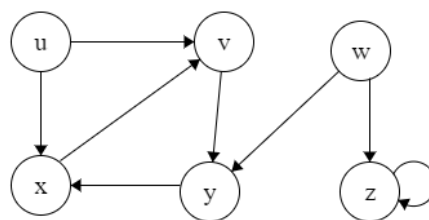
- 13 a) Give Breadth First Search algorithm for graph traversal. Perform its complexity analysis. (7)

- b) Define AVL tree. Construct an AVL tree by inserting the keys: 44, 17, 32, 78, 50, 88, 48, 62, 54 into an initially empty tree. Write clearly the type of rotation performed at the time of each insertion. (7)

OR

- 14 a) Give Depth First Search algorithm for graph traversal. Perform its time complexity analysis. (7)

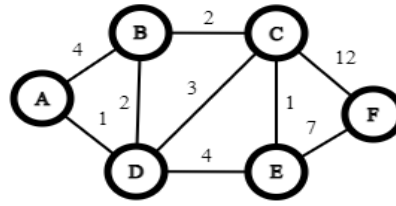
- b) Perform DFS traversal on the following graph starting from node A. When multiple nodes are available for next traversal choose nodes in alphabetical order. Classify the edges of the graph into different category. (7)



Module III

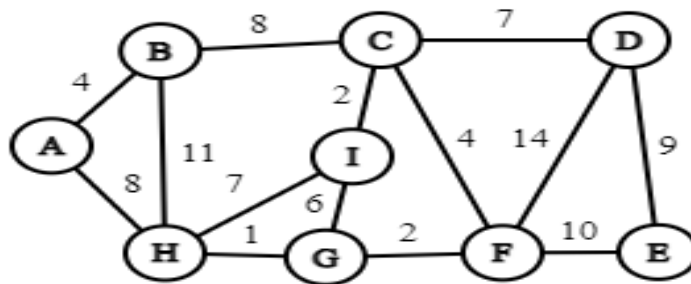
- 15 a) Illustrate the divide and conquer approach by applying 2 way merge sort for the input array: [15,12,14,17,11,13,12,16]. Write the recurrence for merge sort and give the complexity. (7)

- b) Apply Dijkstra's algorithm for single source shortest path to solve the following graph. Assume the source as node A. (7)



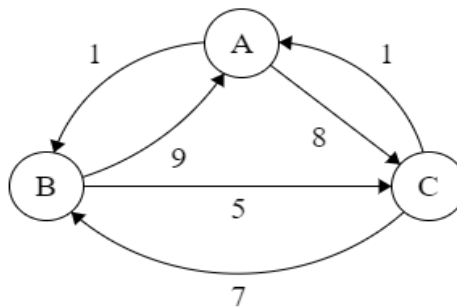
OR

- 16 a) Consider the following instance of Fractional Knapsack problem with 3 objects. (7)
 The capacity of the knapsack is 20 units. The weights and profits of the 3 items respectively are represented by the vectors $(w_1, w_2, w_3) = (18, 15, 10)$ and $(p_1, p_2, p_3) = (25, 24, 15)$. Using a greedy strategy compute the optimal solution to this instance.
- b) Apply Kruskal algorithm to find minimum cost spanning tree for the following (7)
 graph



Module IV

- 17 a) Discuss Floyd-Warshall algorithm for all pair shortest path problem. Solve the (8)
 following instance using the algorithm.

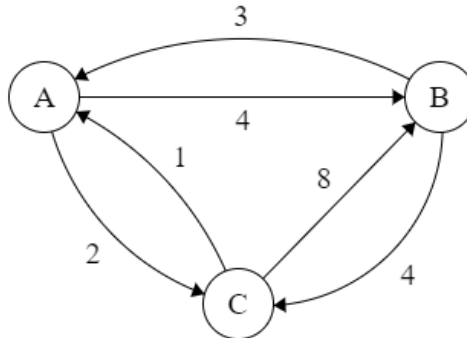


- b) Discuss the control abstraction used in backtracking design technique. Draw the (6)
 state space tree for 4-queens problem.

OR

- 18 a) Discuss the elements of dynamic programming by considering the matrix chain (5)
 multiplication problem.

- b) Define Travelling Salesman Problem (TSP). Apply branch and bound technique to solve the following instance of TSP. Assume that the starting vertex as A. Draw the state space tree for each step. (9)



Module V

- 19 a) Prove that CLIQUE problem is NP Complete. (7)
 b) Write randomized quicksort algorithm and perform its expected running time analysis. (7)

OR

- 20 a) Define approximation algorithm. Give an approximation algorithm for bin packing using first fit heuristic and give its approximation ratio. (7)
 b) Discuss the advantages of randomized algorithms over deterministic algorithms. (7)
 Discuss Las Vegas and Monte Carlo algorithms with a suitable example.
